

Applied Machine Learning — Assignment 04

PART A — CNN PLANT DISEASE CLASSIFICATION REPORT

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This report presents the results and observations obtained from building and training a Convolutional Neural Network (CNN) for plant disease classification using the PlantVillage dataset. The objective of the task was to classify plant leaf images into 38 disease and healthy categories with high accuracy.

Dataset Summary

Metric	Value
Total Classes	38
Training Images	43,444
Validation Images	10,861
Image Resolution	128 × 128
Batch Size	32
Best Validation Accuracy	96.30%
Validation Loss	0.1408
Best Epoch	15

Peach__healthy



Tomato__Spider_mites Two-spotted_spider_mite



Grape__Leaf_blight_(Isariopsis_Leaf_Spot)



Pepper,_bell__Bacterial_spot



Orange__Haunglongbing_(Citrus_greening)



Peach__Bacterial_spot



Potato__Late_blight

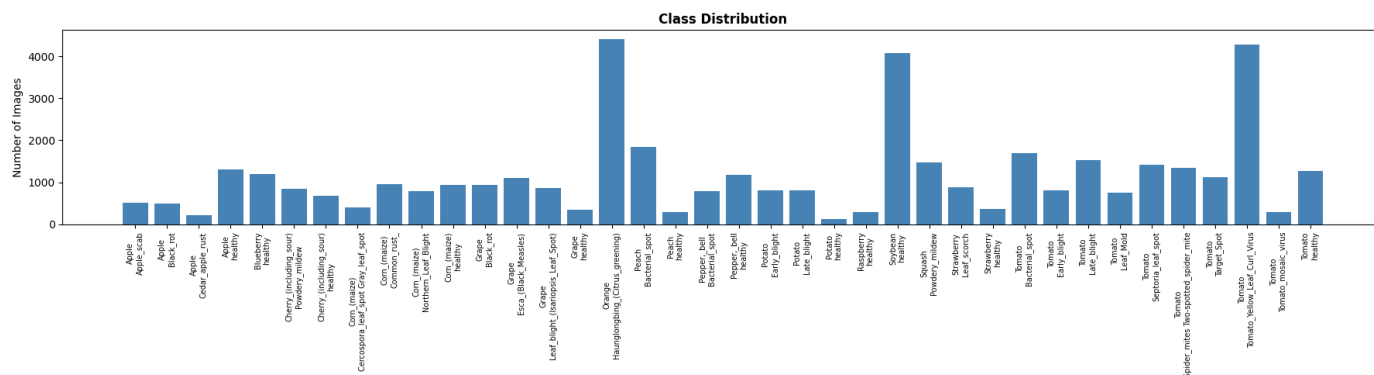


Tomato__Septoria_leaf_spot



Cherry_(including_sour)__Powdery_mildew





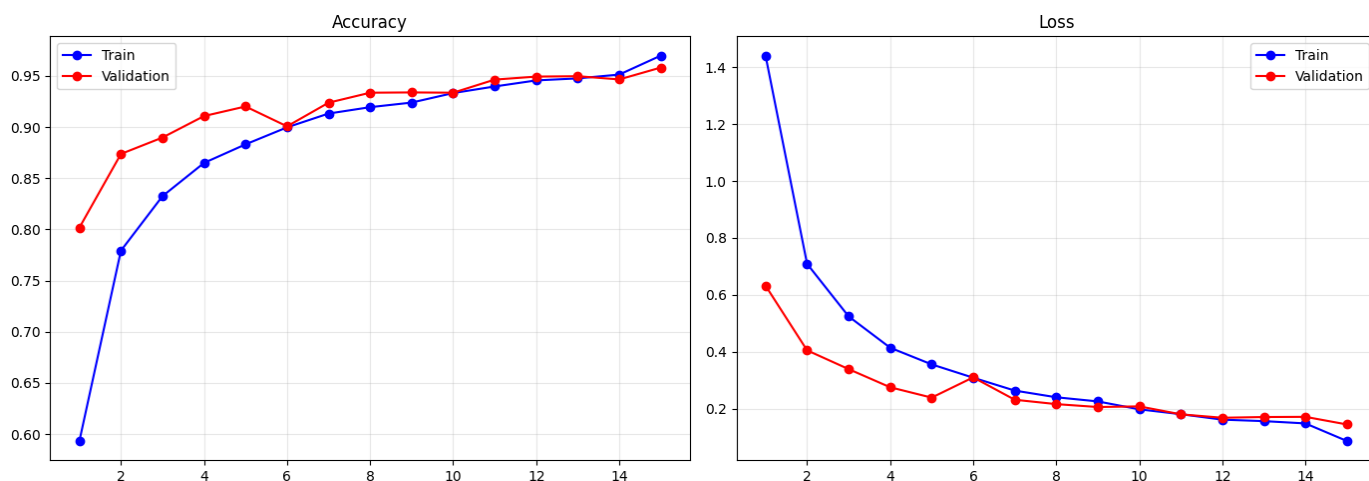
Training Results

The CNN model showed strong learning performance throughout training. Validation accuracy improved steadily from approximately 80% during the first epoch to 96.30% by the final epoch. Simultaneously, validation loss continuously decreased, indicating that the model generalized effectively on unseen data. The training curves showed stable convergence without severe overfitting. The model successfully learned disease-related visual patterns from plant leaf images and achieved highly accurate predictions across most classes.

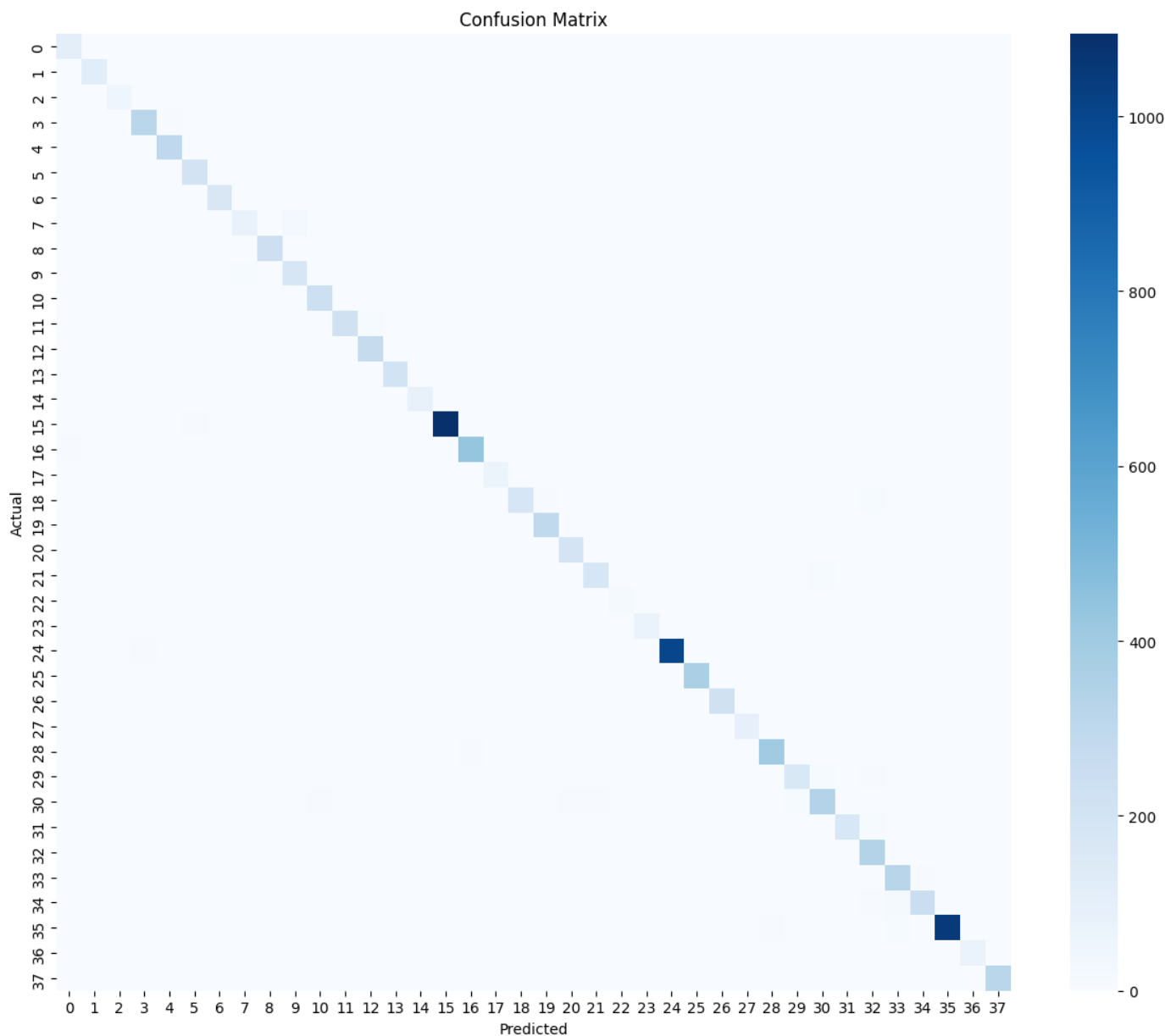
Optimizer and Learning Rate Strategy

The model used the **Adam optimizer** with an initial learning rate of **0.001**. Adam was selected because it combines the advantages of Momentum and RMSProp optimization techniques, allowing faster convergence and more stable weight updates during training. It is highly effective for deep learning models and performs well on large image datasets. A **ReduceLROnPlateau** callback dynamically reduced the learning rate whenever validation performance stopped improving. During training, the learning rate decreased from **0.001** to **0.0005** and later to **0.00025**. This adaptive strategy helped the model fine-tune its learning process, avoid unstable updates, and achieve improved validation accuracy.

Training Rate:



Confusion matrix:



Performance Analysis

The model achieved excellent classification performance across most disease categories. Several classes obtained near-perfect precision, recall, and F1-scores. Particularly strong performance was observed in:

- Corn Common Rust
- Soybean Healthy
- Tomato Yellow Leaf Curl Virus
- Squash Powdery Mildew
- Cherry Powdery Mildew

The overall weighted F1-score reached approximately 96%, demonstrating reliable performance across the entire validation dataset.



Challenges and Observations

Although the model performed very well overall, a few disease categories showed comparatively lower recall and precision values. These included:

- Corn Cercospora Leaf Spot

- Tomato Early Blight

- Potato Healthy

- Tomato Late Blight

These categories contain visually similar symptoms, which increased classification difficulty. Additionally, some classes had fewer training samples, which may have contributed to minor prediction errors.

Key Learnings

This experiment demonstrated the effectiveness of CNNs for agricultural image classification tasks. The model successfully learned disease-specific patterns directly from images without transfer learning. Key learnings from the project include:

- CNN architectures are highly effective for feature extraction from plant images.
- Dropout layers and learning-rate scheduling improve generalization.
- Balanced datasets significantly improve classification performance.
- Proper optimization strategies such as Adam improve convergence speed and stability.

Conclusion

The developed CNN model achieved a final validation accuracy of **96.30%** on the PlantVillage dataset. The model demonstrated strong classification capability across 38 plant disease categories and showed stable convergence during training. Overall, the experiment highlights the practical potential of deep learning systems for automated plant disease detection and agricultural support applications.

Part B: LSTM-based Sentiment Analysis Report

This report summarizes the results, observations, evaluation metrics, and findings obtained from the implementation of a custom LSTM-based Recurrent Neural Network (RNN) for IMDb movie review sentiment classification. The objective of the task was to classify reviews into positive and negative sentiments using sequential deep learning techniques.

Dataset and Preprocessing

The IMDb Large Movie Review Dataset containing 50,000 labeled reviews was used for binary sentiment classification. All reviews were manually preprocessed by converting text to lowercase, removing HTML tags, punctuation, and stop words. A custom vocabulary of the 20,000 most frequent words was created, and reviews were converted into integer token sequences. All sequences were padded or truncated to a fixed length of 300 tokens. The dataset was manually split into 80% training and 20% validation samples.

LSTM Architecture Decisions

The model used an Embedding layer with trainable embeddings of size 64 followed by two stacked LSTM layers. The first LSTM layer contained 128 hidden units with `return_sequences=True`, while the second LSTM layer contained 64 hidden units. Batch normalization was placed between recurrent layers to stabilize training and improve convergence. Dropout layers were included to reduce overfitting and improve generalization. A dense layer with ReLU activation was used before the final sigmoid output neuron for binary classification.

Training Results

The model was trained for 15 epochs using the Adam optimizer and binary cross-entropy loss. Early stopping and ReduceLROnPlateau were applied to improve convergence and reduce overfitting. Training initially progressed slowly during the early epochs, but performance improved significantly after learning rate reduction. The best validation accuracy achieved was **88.75%** at epoch 15.

Metric	Value
Validation Accuracy	0.8875

Precision	0.8855
Recall	0.8910
F1-Score	0.8882
AUC Score	0.9555

Confusion Matrix Analysis

The confusion matrix demonstrated strong classification performance with balanced predictions across both sentiment classes. The model correctly classified 4,471 positive reviews and 4,404 negative reviews. Misclassifications mainly occurred for reviews containing mixed or neutral language, where both positive and negative expressions appeared together.

ROC Curve and AUC

The ROC curve showed strong separability between positive and negative sentiments. An AUC score of 0.9555 indicates that the model achieved excellent discrimination capability and was highly effective in distinguishing sentiment classes.

Baseline Comparison

A majority-class baseline achieved approximately 49.82% accuracy, which is close to random guessing because the dataset is balanced. The LSTM model significantly outperformed the baseline with 88.75% validation accuracy, demonstrating the effectiveness of sequential learning for sentiment analysis.

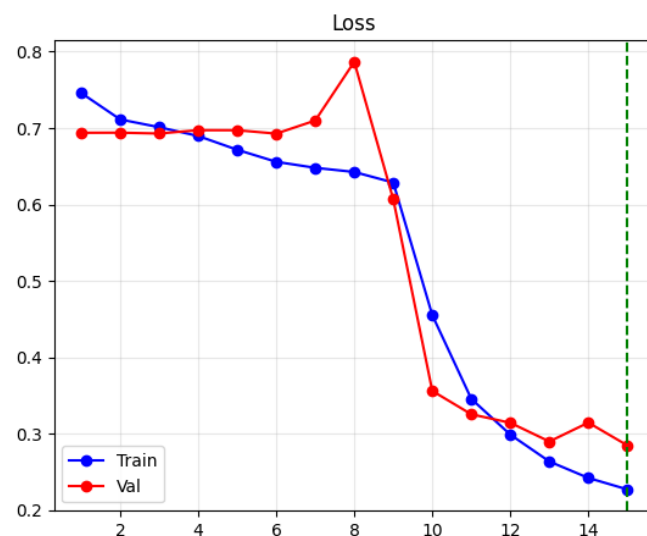
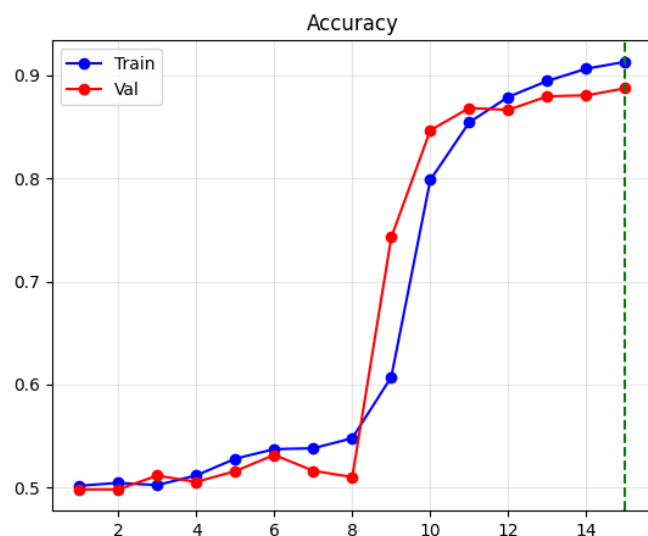
Challenges and Observations

One challenge observed during training was unstable validation performance during early epochs. This issue was addressed using learning rate scheduling, dropout regularization, and batch normalization. The model also faced difficulty with reviews containing sarcasm, mixed emotions, or ambiguous language. Despite these challenges, the model generalized well and achieved strong overall performance.

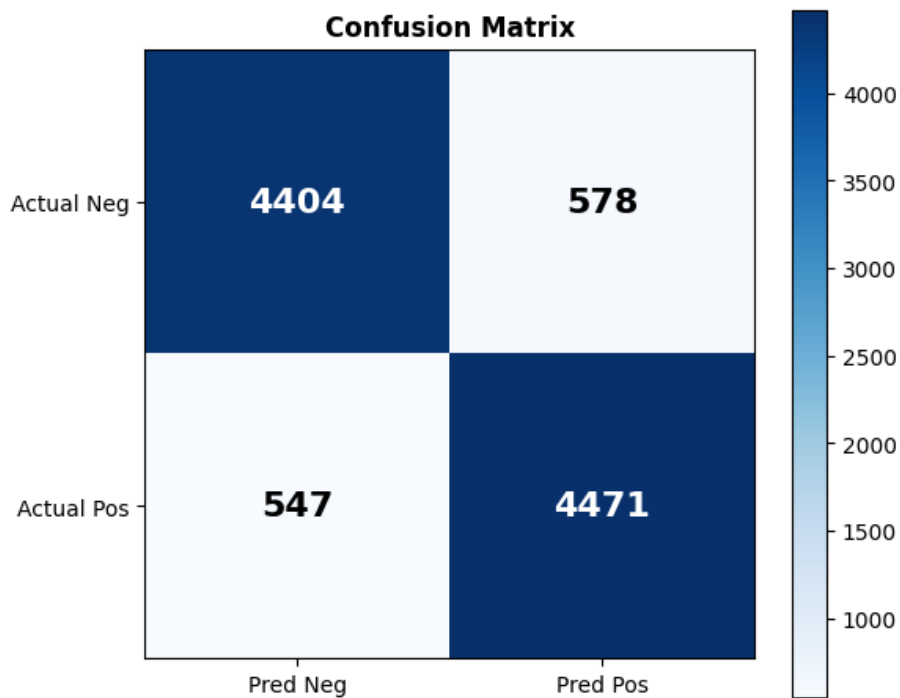
Handling Long-Range Dependencies

The stacked LSTM architecture enabled the network to capture sequential dependencies and contextual relationships across long reviews. Unlike traditional feedforward approaches, the LSTM retained important contextual information over time using gating mechanisms, which improved sentiment understanding for lengthy reviews.

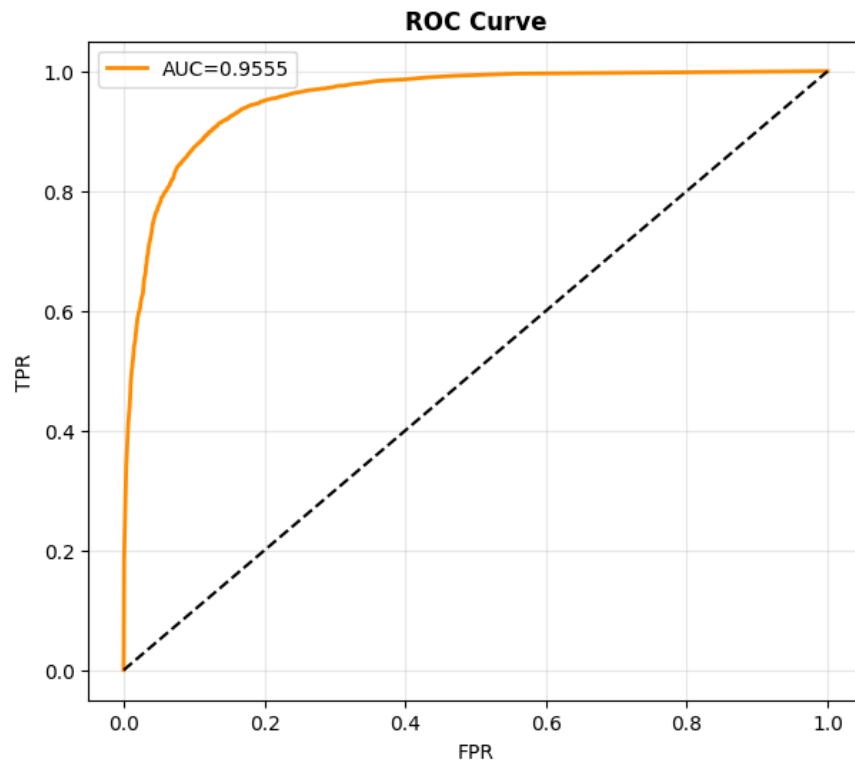
Training and Validation Curves



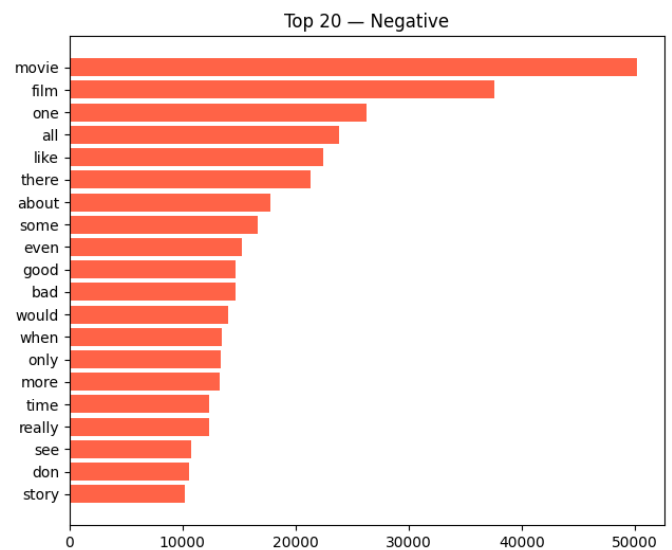
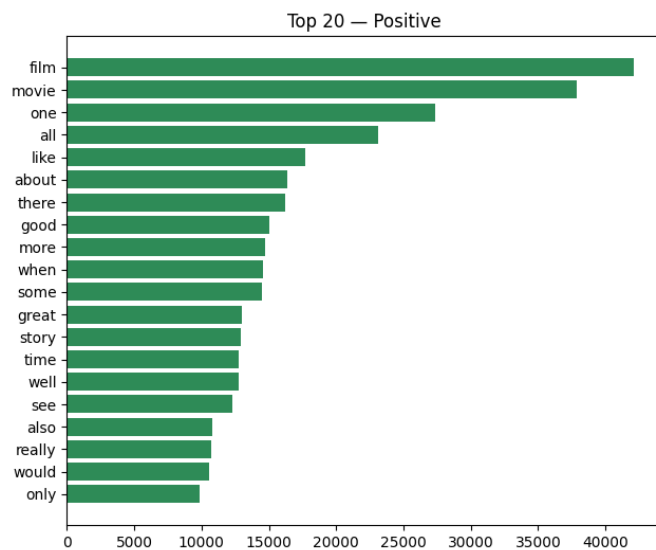
Confusion Matrix



ROC Curve



Top Word Frequencies



Conclusion

The custom LSTM-based sentiment analysis system achieved strong performance on the IMDb dataset with high accuracy, precision, recall, and AUC score. The results demonstrate the capability of recurrent neural networks to model contextual information and sequential dependencies in natural language. Regularization techniques and adaptive learning rate scheduling helped reduce overfitting and improve generalization performance.